

Measured NLP Solve Time

When the sim was started

1st cold start $\rightarrow$ 320ms

Warmed start times $\rightarrow$ 8ms

Once stable $\rightarrow$ 4ms

Our simulation timestep we have chosen as 50Hz

$\Rightarrow$ 20ms

$\Rightarrow$ In this 20ms, the solver picks up the real signal, solves for costs, etc and finds optimal settings $\rightarrow$ If the solve time exceeds the 20ms $\Rightarrow$ Leads to a delay as new real input is available but the output is logged.

Our solve time of 8ms is well within the 20ms window, and enough buffer is there for accounting for latency, etc.

This 20 ms window is not more than it
because then the delta between the inputs
increases

The RK4 assumes rather simple equations within
a timestep, and hence increasing it alot leads
to worsened accuracy.

τ is constant in a certain ' dt ' as well